

Long-range order and spontaneous symmetry breaking in quantum spin systems and hard-core bosons on a lattice

part 5 ODLRO and SSB in the ground states of hard-core bosons on a lattice

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

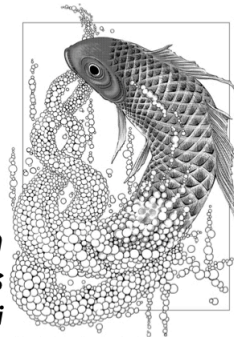
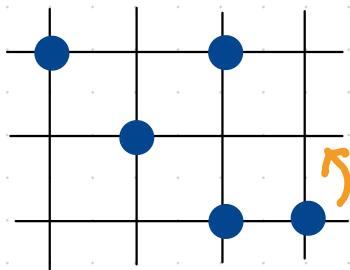


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§ setting and the Hamiltonian

atoms



system of bosons trapped on sites of Λ_L

hard-core: at most one particle on each site

bosons can hop (tunnel) to neighboring sites

an idealized model of ultra-cold bosonic atoms in an optical lattice

▷ state at site $u \in \Lambda_L$ $|0\rangle_u$ no particles $|1\rangle_u$ one particle

▷ particle configuration $|n\rangle = (n_u)_{u \in \Lambda_L}$ with $n_u = 0, 1$ (1)

▷ corresponding basis state $|n\rangle = \bigotimes_{u \in \Lambda_L} |n_u\rangle_u$ (2)

▷ Hilbert spaces $\mathcal{H}_L = \text{span}\{|n\rangle \mid \text{all possible } n\}$ (3)

↙ $\mathcal{H}_{L,N} = \text{span}\{|n\rangle \mid \text{all } n \text{ s.t. } \sum_u n_u = N\}$ (4)

physical Hilbert space for isolated atoms
(not photons!)

↗ Fock space $\mathcal{H}_L = \bigoplus_{N=0}^{L^d} \mathcal{H}_{L,N}$ (5)

▷ basic operators $\begin{cases} \hat{a}_u^\dagger \text{ creation operator for boson at } u \\ \hat{a}_u \text{ annihilation operator for boson at } u \end{cases}$

$$\hat{a}_u^\dagger |0\rangle_u = |1\rangle_u, \hat{a}_u^\dagger |1\rangle = 0, \hat{a}_u |0\rangle_u = 0, \hat{a}_u |1\rangle_u = |0\rangle_u \quad (1)$$

number operator $\hat{n}_u = \hat{a}_u^\dagger \hat{a}_u \quad (2) \quad \hat{n}_u |0\rangle_u = 0, \hat{n}_u |1\rangle_u = |1\rangle_u \quad (3)$

$$\hat{N}_L = \sum_{u \in \Lambda_L} \hat{n}_u \quad (4) \quad \hat{N}_L |\Phi\rangle = N |\Phi\rangle \text{ for } |\Phi\rangle \in \mathcal{H}_{L,N} \quad (5)$$

Hamiltonian $\hat{H}_L = -\frac{1}{2} \sum_{\{u,v\} \in \mathcal{B}_L} \{\hat{a}_u^\dagger \hat{a}_v + \hat{a}_v^\dagger \hat{a}_u\} \quad (6)$

mandatory!
symmetry

$$[\hat{H}_L, \hat{N}_L] = 0 \quad (7)$$

U(1) symmetry

$$e^{i\theta \hat{N}_L} \hat{H}_L e^{-i\theta \hat{N}_L} = \hat{H}_L \quad (8)$$

▷ remark: bosonic Hubbard model

$$\hat{b}_u, \hat{b}_u^\dagger \text{ standard boson operators } [\hat{b}_u, \hat{b}_v^\dagger] = \delta_{u,v}, [\hat{b}_u, \hat{b}_v] = 0 \quad (9)$$

$\hat{n}_u = \hat{b}_u^\dagger \hat{b}_u$ has eigenvalues $0, 1, 2, \dots$

$$\hat{H}_L^{\text{b.H.}} = -\frac{1}{2} \sum_{\{u,v\} \in \mathcal{B}_L} \{\hat{b}_u^\dagger \hat{b}_v + \hat{b}_v^\dagger \hat{b}_u\} + U \sum_{u \in \Lambda_L} \hat{n}_u (\hat{n}_u - 1) \quad (10)$$

$U \uparrow \infty$ limit $\Rightarrow$ hard-core bosons $\hat{n}_u (\hat{n}_u - 1) = 0$ if $n_u = 0, 1$

§ equivalence to the XY model

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states $|0\rangle_u \leftrightarrow |\downarrow\rangle_u$ $|1\rangle_u \leftrightarrow |\uparrow\rangle_u$ (1)

operators $\hat{a}_u^\dagger \leftrightarrow \hat{S}_u^+$ $\hat{a}_u \leftrightarrow \hat{S}_u^-$ $\hat{n}_u \leftrightarrow \hat{S}_u^{(z)} + \frac{1}{2}$ (2)

$\hat{H}_L^{\text{hard core boson}} \leftrightarrow \hat{H}_L^{\text{XY}}$ (3)

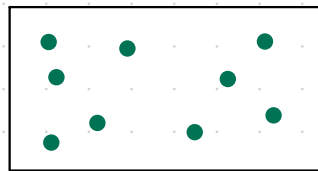
mathematically the same model in different languages

▶ physically relevant Hilbert spaces essential difference!!

XY model: $\mathcal{H}_L$ $\hat{O}_L^{(z)} = \sum_u \hat{S}_u^{(z)}$ is a conserved quantity
but the conservation law is not exact in a real magnetic system

$|\Phi_L^{\text{GS}}\rangle$ the lowest energy state in whole $\mathcal{H}_L$

hard core bosons: $\mathcal{H}_{L,N}$ with a fixed particle number N



particle number conservation is an exact law for an isolated atomic system

(not for photons!)

$|\Phi_{L,N}^{\text{GS}}\rangle$ the lowest energy state in each $\mathcal{H}_{L,N}$

§ ODLRO in the ground state of the half-filled model

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$|\Phi_{L,N}^{GS}\rangle$ ground state for the model on Λ_L with N particles

$$\hat{O}_L^+ = \sum_{u \in \Lambda_L} \hat{a}_u^\dagger, \quad \hat{O}_L^- = \sum_{u \in \Lambda_L} \hat{a}_u \quad (1)$$

clearly $\langle \Phi_{L,N}^{GS} | \hat{O}_L^+ | \Phi_{L,N}^{GS} \rangle = 0 \quad (2)$

→ follows from Mattis 1979, Kennedy, Lieb, Shastry 1988, Kubo, Kishi 1988

Theorem for $N = L^d/2$, the ground state $|\Phi_{L,N}^{GS}\rangle$ is unique
for $d=2,3,\dots$, there is a (d -dependent) constant $\varrho_0 > 0$ s.t. for any L

$$\langle \Phi_{L,N}^{GS} | \frac{\hat{O}_L^+ \hat{O}_L^-}{L^{2d}} | \Phi_{L,N}^{GS} \rangle = \langle \Phi_{L,N}^{GS} | \frac{\hat{O}_L^- \hat{O}_L^+}{L^{2d}} | \Phi_{L,N}^{GS} \rangle \geq 2\varrho_0 \quad (3)$$

(proof rewrite the theorems for the XY model (part 4, p3) with

$$\hat{O}_L^{(X)} = \frac{1}{2}(\hat{O}_L^+ + \hat{O}_L^-), \quad \hat{O}_L^{(Y)} = \frac{1}{2i}(\hat{O}_L^+ - \hat{O}_L^-) \quad (4)$$

$$\langle \Phi_{L,N}^{GS} | \frac{\hat{\sigma}_L^+ \hat{\sigma}_L^-}{L^{2d}} | \Phi_{L,N}^{GS} \rangle = \langle \Phi_{L,N}^{GS} | \frac{\hat{\sigma}_L^- \hat{\sigma}_L^+}{L^{2d}} | \Phi_{L,N}^{GS} \rangle \geq 2\varrho_0 \quad (1) \quad 5$$

establishes the existence of Bose-Einstein condensation (BEC)
in terms of off-diagonal long-range order (ODLRO)

(1) suggests $\rightarrow \langle \Phi_{L,N}^{GS} | \hat{a}_u^\dagger \hat{a}_v | \Phi_{L,N}^{GS} \rangle \geq 2\varrho_0 > 0$ for large $|u-v|$ (2)

ODLRO fundamental characterization of BEC and superconductivity

Penrose 1951, Penrose, Onsager 1956, Yang 1962

$\rightarrow$ for the traditional treatment without dynamical EM field

standard LRO $\hat{S}_u \cdot \hat{S}_v$



ODLRO $\hat{a}_u^\dagger \hat{a}_v$

$\leftarrow$ create
u

$\leftarrow$ annihilate
v

ODLRO is expected for other $p = N/L^d$, but is proved only for $p = 1/2$ $\rightarrow$ we don't know the right proof!

(ODLRO is obvious in the ground state of a free boson model ($U=0$ in p2-(10))

$$| \Phi_{L,N}^{GS,0} \rangle = (\text{const.}) \left(\sum_u \hat{b}_u^\dagger \right)^N \left(\otimes_u |0\rangle_u \right) \quad (3) \quad \langle \Phi_{L,N}^{GS,0} | \hat{b}_u^\dagger \hat{b}_v | \Phi_{L,N}^{GS,0} \rangle = N/L^d \quad (4)$$

Slow-lying states

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for hard core bosons with $N = L^d/2$ and $d \geq 2$, Koma-Tasaki theory readily extends

$$\underline{|\Gamma_{L,N}^M\rangle} = \frac{(\hat{O}_L^+)^M |\Phi_{L,N}^{\text{GS}}\rangle}{\|(\hat{O}_L^+)^M |\Phi_{L,N}^{\text{GS}}\rangle\|}, \quad \underline{|\Gamma_L^{-M}\rangle} = \frac{(\hat{O}_L^-)^M |\Phi_{L,N}^{\text{GS}}\rangle}{\|(\hat{O}_L^-)^M |\Phi_{L,N}^{\text{GS}}\rangle\|} \quad \text{for } M=1, 2, \dots \quad (1)$$

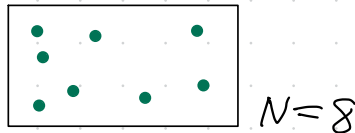
Theorem (Koma, Tasaki 1994)

part 4 p10

for $d=2, 3, \dots$ there are (d -dependent) constants C_1, C_2 s.t.

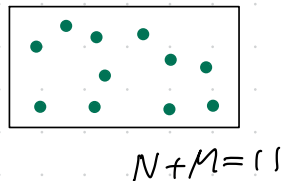
$$\underline{0 \leq \langle \Gamma_{L,N}^M | \hat{H}_L | \Gamma_{L,N}^M \rangle - E_{L,N}^{\text{GS}} \leq C_1 \frac{M^2}{L^d}} \quad (2)$$

for any M with $|M| \leq C_2 L^{d/2}$ and any L



note however that $|\Gamma_{L,N}^M\rangle \in \mathcal{H}_{L,N+M}$ (3)

a different Hilbert space...



similarly for the exact excited state $|\Psi_{L,N}^M\rangle$ in corollary (part 4 - p10)

§ low-lying states with full SSB

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$$|\Xi_{L,N}^\theta\rangle = \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} e^{-i\theta M} |\Xi_{L,N}^M\rangle \in \mathcal{H}_L = \bigoplus_{N'=0}^{L^d} \mathcal{H}_{L,N'} \quad (1)$$

the "macroscopic ground state" in the Fock space $\mathcal{H}_L$ where

$(\hat{O}_L^{(x)}/L^d, \hat{O}_L^{(y)}/L^d)$ have definite values $(m^* \cos \theta, m^* \sin \theta)$

$\hat{O}_L^\pm/L^d$ have definite values $m^* e^{\pm i\theta}$

in particular $\langle \Xi_{L,N}^\theta | \hat{a}_u^\dagger | \Xi_{L,N}^\theta \rangle \simeq m^* e^{i\theta} \quad (2)$

$$\langle \Xi_{L,N}^\theta | \hat{a}_u | \Xi_{L,N}^\theta \rangle \simeq m^* e^{-i\theta} \quad (3)$$

the creation/annihilation operators have nonzero expectation values!

$|\Xi_{L,N}^\theta\rangle$ fully breaks the $U(1)$ symmetry, but it is a superposition of states with different particle numbers

NOT ALLOWED!!

in $\mathcal{H}_{L,N}$

▶ can the environment help? **NO**

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$$\mathcal{H}_{\text{whole}} = \mathcal{H}_{\text{system}} \otimes \mathcal{H}_{\text{env}} \quad (1)$$

N_{tot} : the total number of bosons

$$\mathcal{H}_{\text{whole}} \supset \sum_N |\Phi_{N \text{ particles}}^{\text{system}} \rangle |\Phi_{N_{\text{tot}} - N \text{ particles}}^{\text{env.}} \rangle \quad (2)$$

tracing out the environment gives

$$\text{a mixed state} \quad \sum_N (\dots) |\Phi_{N \text{ particles}}^{\text{system}} \rangle \langle \Phi_{N \text{ particles}}^{\text{system}}| \quad (3)$$

$$\text{rather than a superposition} \quad \sum_N (\dots) |\Phi_{N \text{ particles}}^{\text{system}} \rangle \quad (4)$$

if one starts with a state with a fixed number of bosons (in one's lab)
then it is strictly impossible to generate a state like $|\vec{\square}_{L,N}^{\theta}\rangle$
that explicitly breaks the $U(1)$ symmetry

§ relations between $|\Phi_{L,N}^{GS}\rangle$ and $|\square_{L,N}^\theta\rangle$ ($N=L^d/2$)

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$$\uparrow |\Phi_{L,N}^{GS}\rangle = \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta |\square_{L,N}^\theta\rangle \quad (1)$$



Schrödinger's cat that CANNOT collapse! because $|\square_{L,N}^\theta\rangle$ is NOT allowed by nature
and is hence realistic

nevertheless...

any physically observable operator $\hat{A}$ satisfies $[\hat{A}, \hat{N}_L] = 0$ (2)

then $\langle \square_{L,N}^\theta | \hat{A} | \square_{L,N}^\theta \rangle$ does not depend on θ

part 4 - p15

we conjecture $\langle \square_{L,N}^\theta | \hat{A} | \square_{L,N}^\theta \rangle \simeq \langle \Phi_{L,N}^{GS} | \hat{A} | \Phi_{L,N}^{GS} \rangle$ for local $\hat{A}$ (2)

↓
one may use $|\square_{L,N}^\theta\rangle$ instead of $|\Phi_{L,N}^{GS}\rangle$ for theoretical calculations!!

mean-field theories

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$|\Xi_{L,N}^{\theta}\rangle$ exhibits both ODLRO and SSB and is theoretically "natural"

mean-field and variational calculations often lead us to states like $|\Xi_{L,N}^{\theta}\rangle$

- Bogoliubov's theory of BEC (Bogoliubov 1947) $\sim \sum_M e^{i\theta M} |\Gamma_{L,N}^M\rangle$
- BCS theory for superconductivity (Bardeen, Cooper, Schieffer 1957)

c-number substitution $\hat{O}_L^{\pm} \rightarrow L^d m^* e^{\pm i\theta} \rightarrow p.7$

- These theories are practically useful because of the expected P9-(2)
- The absolute global phase θ is unobservable (relative phases may be measured)

Common confusions: Q and A

- can BEC/superconductivity take place in a system with a fixed number of particles?
 - yes! $|\Phi_{L,N}^{GS}\rangle$ is in $\mathcal{H}_{L,N}$ and perfectly exhibits BEC !!
- does the existence of BEC/superconductivity induce particle number fluctuation?
 - not necessarily! The superposition in $|\Xi_{L,N}^{\theta}\rangle$ is NOT directly related to physically observable particle number fluctuation

§ SSB in coupled BECs

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$$p7-(1) \quad |\Box_{L,N}^\theta\rangle = \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} e^{-i\theta M} |\Gamma_{L,N}^M\rangle$$

$$p9-(1) \quad |\Phi_{L,N}^{GS}\rangle = \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta |\Box_{L,N}^\theta\rangle$$

two identical copies of the g.s. of the hard core bosons on Λ_L and $N=L^d/2$

$$\begin{aligned} |\Phi_{L,N}^{GS}\rangle_1 |\Phi_{L,N}^{GS}\rangle_2 &= \frac{2M(L)+1}{(2\pi)^2} \int_0^{2\pi} d\theta_1 \int_0^{2\pi} d\theta_2 |\Box_{L,N}^{\theta_1}\rangle_1 |\Box_{L,N}^{\theta_2}\rangle_2 \\ &= \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\varphi \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta |\Box_{L,N}^\theta\rangle_1 |\Box_{L,N}^{\theta+\varphi}\rangle_2 \end{aligned}$$

$\equiv |\Omega_{L,N}^\varphi\rangle_{12}$

with

$$\begin{aligned} |\Omega_{L,N}^\varphi\rangle_{12} &= \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta \frac{1}{2M(L)+1} \sum_{M=-M(L)}^{M(L)} e^{-i\theta M} |\Gamma_{L,N}^M\rangle_1 \sum_{M'=-M(L)}^{M(L)} e^{-i(\theta+\varphi)M'} |\Gamma_{L,N}^{M'}\rangle_2 \\ &= \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} e^{iM\varphi} |\Gamma_{L,N}^M\rangle_1 |\Gamma_{L,N}^{-M}\rangle_2 \end{aligned}$$

→ total particle number is $2N$

We thus see

$$|\Phi_{L,N}^{GS}\rangle_1 |\Phi_{L,N}^{GS}\rangle_2 = \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\varphi |\Omega_{L,N}^\varphi\rangle_{12} \quad (1)$$



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$$\text{with } |\Omega_{L,N}^\varphi\rangle_{12} \in \bigoplus_{M=-N}^N \mathcal{H}_{L,N+M}^{(1)} \otimes \mathcal{H}_{L,N-M}^{(2)} \quad (2) \quad (N=L^d/2)$$

almost like weak
magnetic field in spin systems
see my paper

physically allowed Hilbert space when the two subsystems
can exchange particles (through a "Josephson junction")

a weak tunneling between the two subsystems may lead to a ground state

$$|\Omega_{L,N}^\varphi\rangle_{12} = \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta |\square_{L,N}^\theta\rangle_1 |\square_{L,N}^{\theta+\varphi}\rangle_2 \quad (3) \quad \text{with some } \varphi$$

relative phase φ between the two condensates is fixed $\rightarrow$ SSB!

while the global phase θ is NOT fixed $\rightarrow$ particle number conservation

(this essentially describes what is realized experimentally)